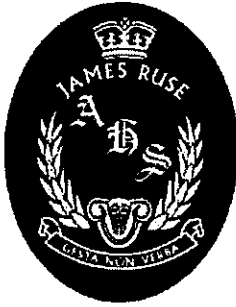


Student Name: \_\_\_\_\_

Maths class: \_\_\_\_\_



James Ruse Agricultural High School

**2020**

YEAR 12 Task 3

# Mathematics Extension 1

## General Instructions

### Reading time – 5 minutes

- Working time – 1 hour and 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided

**Total Marks: 63**

Attempt all questions (1 – 7). Each question is worth 9 marks.

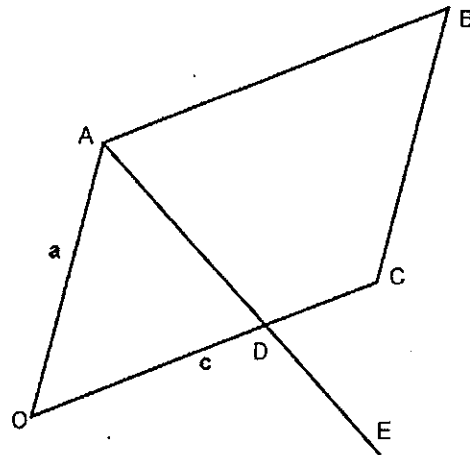
Start your answer to each question on a separate piece of paper.

**Question 1 (Start a new piece of paper)**

**Marks**

- (a)  $OABC$  is a parallelogram with  $O$  as the origin. The position vector of  $A$  is  $\underline{a}$  and the position vector of  $C$  is  $\underline{c}$ .  $D$  divides the segment  $OC$  such that  $OD:DC = 2:1$ .  $E$  is a point on  $AD$  extended so that  $AD:DE = 3:2$ .

**4**



(Diagram not to scale)

Find the following vectors in terms of  $\underline{a}$  and  $\underline{c}$ :

- (i)  $\overrightarrow{AC}$       (ii)  $\overrightarrow{AD}$       (iii)  $\overrightarrow{AE}$       (iv)  $\overrightarrow{BE}$
- (b) (i) Explain why the product  $n(n+1)$  is even for any positive integer  $n$ . **1**
- (ii) Prove by Mathematical Induction that  $5n^3 + 7n$  is divisible by 6 for all positive integers  $n$ . **4**

**Question 2 (Start a new piece of paper)**

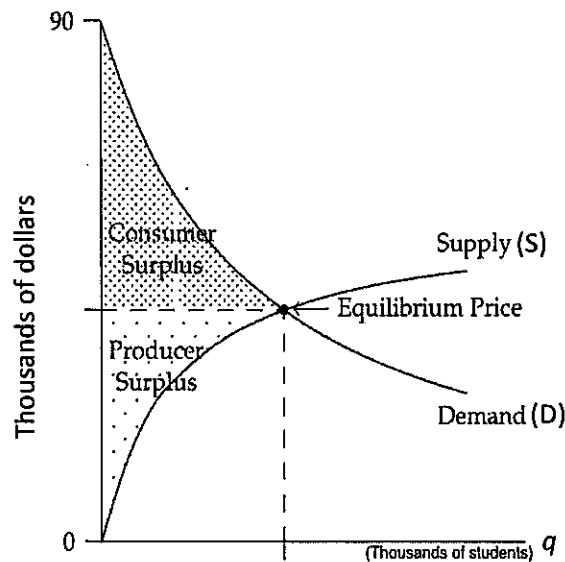
**Marks**

- (a) The area bounded by the curve  $y = \ln(2x)$ , the  $y$  axis and the lines  $y = \ln 2$  and  $y = \ln 6$  is rotated about the  $y$  axis. Find the volume of the solid so formed. **3**
- (b) Consider a parallelogram  $WXYZ$ .
- (i) Explain why  $\overrightarrow{XY} = \overrightarrow{WZ}$  and (similarly)  $\overrightarrow{WX} = \overrightarrow{ZY}$ . **1**
- (ii) If  $\overrightarrow{WX} = \underline{a}$  and  $\overrightarrow{XY} = \underline{b}$ , express the diagonals  $\overrightarrow{WY}$  and  $\overrightarrow{XZ}$  in terms of  $\underline{a}$  and  $\underline{b}$ . **2**
- (iii) Use scalar products to find the square of the length of each diagonal in terms of  $\underline{a}$  and  $\underline{b}$ . **2**
- (iv) Hence deduce that the sum of the squares of the diagonal lengths of a parallelogram is equal to sum of the squares of the four sides. **1**

Question 3 (Start a new piece of paper)

Marks

- (a)  $ABCD$  is a parallelogram where  $A$ ,  $B$  and  $C$  have position vectors  $2\mathbf{i} - 4\mathbf{j}$ ,  $-3\mathbf{i} + 3\mathbf{j}$  and  $\mathbf{i} + 5\mathbf{j}$  respectively, relative to an origin  $O$ . Find the position vector of  $D$ . 2
- (b) The Demand curve for students wanting to attend JR University each year is given by Demand Price ( $D$ ), where  $D = \frac{180}{q+2}$ ,  $q$  is measured in thousands of students and  $D$  is measured in thousands of dollars. The university is willing to accept students according to the formula for Supply Price ( $S$ ), where  $S = \frac{56q}{q+1}$ , as shown in the diagram below.



By showing that, when Demand Price is equal to Supply Price,  $q = 2.5$  (to 2 significant figures), and that the equilibrium price is \$40 000, find the Consumer Surplus, as shaded on the diagram. Give your answer in dollars to 2 significant figures. 4

- (c) In what ratio does the  $x$ -axis divide the region bounded by the parabolas  $y = 4x - x^2$  and  $y = x^2 - x$ ? 3

Question 4 continues over the page

**Question 4 (Start a new piece of paper)****Marks**

- (a) Consider the quadrilateral  $ABCD$  where, relative to an origin  $O$ ,  
 $\overrightarrow{OA} = 3\mathbf{i} - \mathbf{j}$ ,  $\overrightarrow{OB} = 3\mathbf{i} + 2\mathbf{j}$ ,  $\overrightarrow{OC} = 9\mathbf{i} + 5\mathbf{j}$  and  $\overrightarrow{OD} = 6\mathbf{i} - \mathbf{j}$ .
- (i) Find  $\overrightarrow{AC}$  and  $\overrightarrow{BD}$  and verify that these vectors are perpendicular. **2**
- (ii) Find  $|\overrightarrow{AB}|$ ,  $|\overrightarrow{BC}|$ ,  $|\overrightarrow{CD}|$  and  $|\overrightarrow{DA}|$  and hence determine to which family of quadrilaterals  $ABCD$  belongs. **3**
- (b) Prove by Mathematical Induction that, for any positive integer  $n$ , **4**
- $$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

**Question 5 (Start a new piece of paper)****Marks**

- (a) Relative to an origin  $O$ , the points  $A$ ,  $B$  and  $C$  have position vectors  $-5\mathbf{i} + 2\mathbf{j}$ ,  $7\mathbf{i} - 4\mathbf{j}$  and  $3\mathbf{i} + p\mathbf{j}$  respectively where  $p$  is a constant.
- (i) Find the value of  $p$  if angle  $ABC$  is  $90^\circ$ . **3**
- (ii) Find the value of  $p$  if  $A$ ,  $B$  and  $C$  are collinear (in some order). **2**
- (b) The curved surface area  $A$  of a solid of revolution generated by rotating the part of a curve between  $x = a$  and  $x = b$  about the  $x$  axis is given by
- $$\int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$
- (i) The curve  $y = 2\sqrt{x}$  is rotated about the  $x$  axis. Show that the curved surface area generated over the interval  $0 \leq x \leq 8$  is given by **2**
- $$4\pi \int_0^8 \sqrt{x+1} dx$$
- (ii) Hence find the required curved surface area. **2**

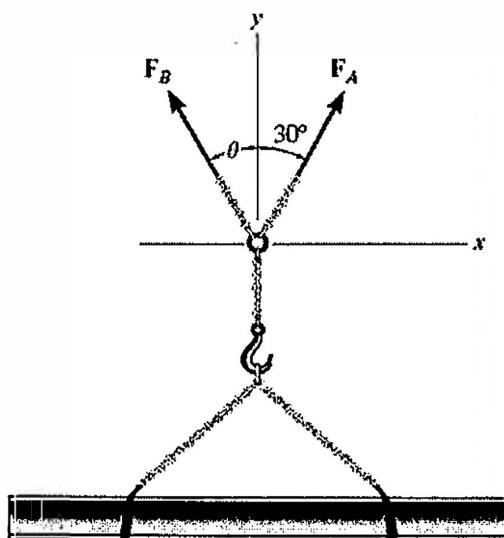
**Question 5 continues over the page**

**Question 6 (Start a new piece of paper)****Marks**

- (a) (i) Find the vector projection of  $15\mathbf{i} + 5\mathbf{j}$  onto  $6\mathbf{i} - 8\mathbf{j}$ . 2
- (ii) Hence break down  $15\mathbf{i} + 5\mathbf{j}$  into components parallel and perpendicular to  $6\mathbf{i} - 8\mathbf{j}$ . 2
- (b) (i) Show that  $(n + 1)^2(3n + 8) = 3n^3 + 14n^2 + 19n + 8$  1
- (ii) Prove by Mathematical Induction that, for any positive integer  $n$ , 4
- $$\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \dots + \frac{1}{n(n+2)} = \frac{n(3n+5)}{4(n+1)(n+2)}$$

**Question 7 (Start a new piece of paper)****Marks**

- (a) The beam shown below is to be hoisted using two chains.



If the resultant force is to be 600N directed along the positive  $y$ -axis,

- (i) Given  $F_A$  acts at  $30^\circ$  from the  $y$ -axis as shown, draw a vector diagram representing the situation and explain why the magnitude of  $F_B$  is a minimum when  $\theta = 60^\circ$ . Hence, 2
- (ii) determine the magnitudes of forces  $F_A$  and  $F_B$  acting on each chain.. 2

Question 7 continues over the page

- (b) (i) The area between the curve  $y = x^2 + 1$ , the  $x$  axis and the ordinates  $x=1$  and  $x=2$  is rotated about the  $x$  axis. Find the volume so formed. 2
- (ii) The area between the curve  $y = x^2 + 1$ , the line  $y = 6$  and the ordinates  $x=1$  and  $x=2$  is rotated about the line  $y = 6$ . 3  
Explain, with the aid of a diagram, why the resulting volume is given by

$$\pi \int_1^2 x^4 - 10x^2 + 25 \, dx$$

(Do not evaluate the integral)

**End of Examination**

**MATHEMATICS Extension 1, Question 1**

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks Awarded                                                                                                                                                                   | Marker's Comments                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(a)</p> <p>(i) <math>\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\underline{a} + \underline{c} = \underline{c} - \underline{a}</math></p> <p>(ii) <math>\overrightarrow{AD} = \overrightarrow{AO} + \overrightarrow{OD} = -\underline{a} + \frac{2}{3}\underline{c} = \frac{2}{3}\underline{c} - \underline{a}</math></p> <p>(iii) <math>\overrightarrow{AE} = \frac{5}{3}\overrightarrow{AD} = \frac{5}{3}\left(\frac{2}{3}\underline{c} - \underline{a}\right) = \frac{10}{9}\underline{c} - \frac{5}{3}\underline{a}</math></p> <p>(iv) <math>\overrightarrow{BE} = \overrightarrow{BA} + \overrightarrow{AE} = -\underline{c} + \frac{5}{3}\overrightarrow{AD} = -\underline{c} + \frac{5}{3}\left(\frac{2}{3}\underline{c} - \underline{a}\right)</math><br/> <math display="block">= \frac{1}{9}\underline{c} - \frac{5}{3}\underline{a}</math></p> <p>(b) (i)<br/> <math>n</math> and <math>(n + 1)</math> are successive integers. Hence one will be even while the other, odd. The product of an odd integer with an even integer yields another even integer. Hence <math>n(n + 1)</math> is even.</p> <p>You could prove it using the definitions of even and odd, those being:</p> <ul style="list-style-type: none"> <li><math>n</math> is even iff there is an integer <math>k</math> such that <math>n = 2k</math></li> <li><math>n</math> is odd iff there is an integer <math>k</math> such that <math>n = 2k + 1</math></li> </ul> <p>Take the product:</p> $n(n + 1) = 2k(2k + 1)$ <p>which is even since <math>k(2k + 1)</math> is an integer given <math>k \in \mathbb{Z}</math>. And if <math>n = 2k + 1</math>, then</p> $\begin{aligned} n(n + 1) &= (2k + 1)(2k + 1 + 1) \\ &= (2k + 1)(2k + 2) \\ &= 2(2k + 1)(k + 1) \end{aligned}$ <p>which is even since <math>(2k + 1)(k + 1)</math> is a combination of products and sums of integers given <math>k \in \mathbb{Z}</math>.</p> | <p>1mark per correct vector.</p> <p>1 mark penalty for lack of vector notation.</p> <p>1 mark awarded for sufficient reasoning. No 'hard' proof required (1 mark question).</p> | <p>Question was handled well generally.</p> <p>Only errors involved were those where direction was not identified correctly or scalar multiples of vectors incorrectly assigned.</p> <p>All students handled this well.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(ii)</p> <p>Let <math>S(n)</math> be the statement that 6 divides <math>5n^3 + 7n</math>; i.e.</p> $6 \mid 5n^3 + 7n \leftrightarrow 5n^3 + 7n = 6m$ <p>for some <math>m \in \mathbb{Z}</math>.</p> <p>We test the base case at <math>n = 1</math>:</p> $5(1)^3 + 7(1) = 12 = 6 \times 2$ <p>The result is a multiple of 6, hence the statement is true for <math>n = 1</math>.</p> <p>Now assume the statement <math>S(n)</math> to be true. Then we consider <math>S(n + 1)</math>:</p> $\begin{aligned} 5(n + 1)^3 + 7(n + 1) &= 5n^3 + 15n^2 + 15n + 5 + 7n + 7 \\ &= (5n^3 + 7n) + 15n^2 + 15n + 12 \\ &= 6m + 15n(n + 1) + 12 \end{aligned}$ <p>(as <math>5n^3 + 7n = 6m</math> by hypothesis).</p> <p>Now, by (i), <math>n(n + 1)</math> is even, hence <math>n(n + 1) = 2K</math> for some <math>K \in \mathbb{Z}</math>. Hence we may continue as</p> $\begin{aligned} 6m + 15n(n + 1) + 12 &= 6m + 15(2K) + 12 \\ &= 6(m + 5K + 2) \end{aligned}$ <p>Since <math>m, K \in \mathbb{Z}</math>, we have that <math>(m + 5K + 2) \in \mathbb{Z}</math>, meaning 6 divides <math>5(n + 1)^3 + 7(n + 1)</math> if 6 divides <math>5n^3 + 7n</math>.</p> <p>Hence if <math>S(n)</math> is true, then <math>S(n + 1)</math> is true.</p> <p>As <math>S(1)</math> is true, it follows by the principle of mathematical induction that <math>S(n)</math> is true for all <math>n \in \mathbb{N}</math>.</p> | <p>1 mark awarded for setup of correct mathematical form and reasonable working from there to the point of the quadratic in <math>S(n + 1)</math>.</p> <p>1 mark for base case.</p> <p>1 mark for dealing appropriately with this section where use of part (i) was made, or a separate induction.</p> <p>1 mark for correct finish, identifying correctly that the LHS of <math>S(n + 1)</math> is a multiple of 6.</p> | <p>Great majority of students established this correctly.</p> <p>All students established the base case.</p> <p>Most students satisfied this part using the result of part (i). Those who didn't use part (i) either used a second induction (fine, but more work) or stopped – can't proceed from this point without showing the quadratic to be divisible by 6.</p> <p>Those who were able to continue needed to then finish appropriately by identifying <math>S(n + 1)</math> as being of form <math>6K</math> for <math>K</math> some integer.</p> <p>This part was finished well.</p> |
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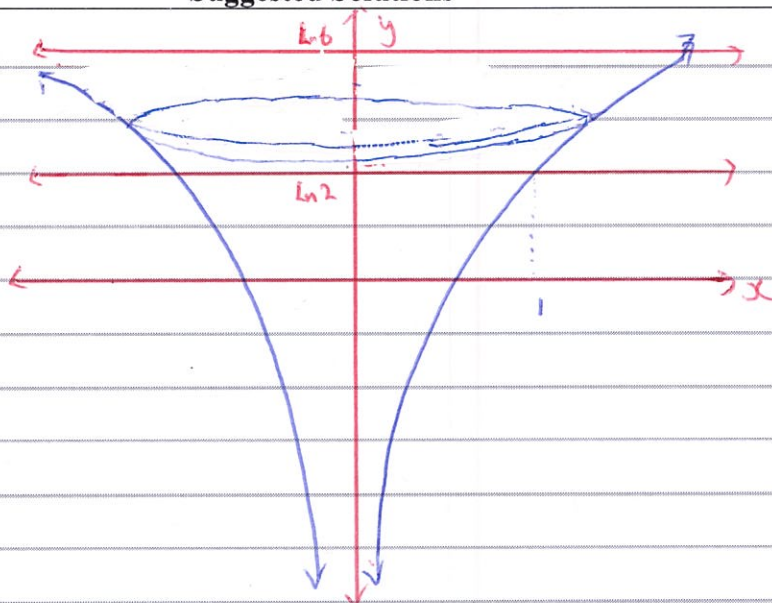
MATHEMATICS Extension 1 : Question...2.....

Suggested Solutions

Marks

Marker's Comments

a)



$$V = \pi \int_{\ln 2}^{\ln 6} x^2 dy$$

$$y = \ln(2x)$$

$$e^y = 2x$$

$$x = \frac{e^y}{2}$$

Write  $x$  in terms of  $y$  ①

$$= \pi \int_{\ln 2}^{\ln 6} \frac{e^{2y}}{4} dy$$

① To Integral

$$= \pi \left[ \frac{e^{2y}}{8} \right]_{\ln 2}^{\ln 6}$$

$$= \pi \left[ \frac{36}{8} - \frac{4}{8} \right]$$

$$= \frac{32}{8} \pi$$

$$= 4\pi$$

① Answer

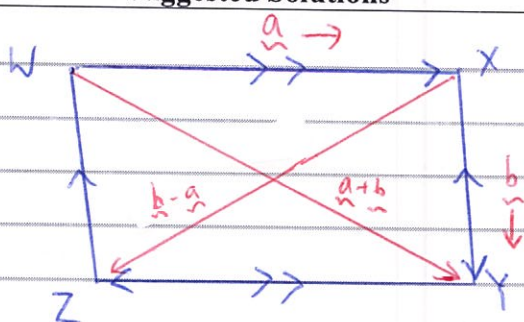
MATHEMATICS Extension 1 : Question. 2.....

Suggested Solutions

Marks

Marker's Comments

b) i)



$\vec{XY} = \vec{WZ}$  because opposite sides of a parallelogram are equal and parallel, and the 2 vectors are in the same direction.  
 $\therefore$  Equal magnitude and direction.

Similarly true for  $\vec{WX}$  and  $\vec{ZY}$ ,  $\therefore \vec{WX} = \vec{ZY}$

ii)  $\vec{WY} = \vec{a} + \vec{b}$

$\vec{XZ} = \vec{b} - \vec{a}$

iii)  $|\vec{WY}|^2 = |\vec{a} + \vec{b}|^2$   
 $= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$   
 $= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$

$|\vec{XZ}|^2 = |\vec{b} - \vec{a}|^2$   
 $= (\vec{b} - \vec{a}) \cdot (\vec{b} - \vec{a})$   
 $= |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{a}|^2$

iv)  $\therefore |\vec{WY}|^2 + |\vec{XZ}|^2 = (|\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2) + (|\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2)$   
 $= 2(|\vec{a}|^2 + |\vec{b}|^2)$

Which is equal to the sum of the squares of the four sides.

Must require either:

- diagram or
- worded explanation about the vertex order so that the vectors that are parallel are not opposites

Students lost marks if you write things such as:

- $a^2 + 2ab + b^2$
- $|a|^2 + 2|a||b| + |b|^2$

Students must either:

- i) write a worded conclusion or

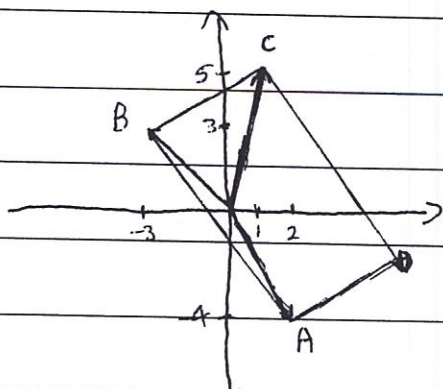
- ii) Connect the expression to  $\vec{WX}$ ,  $\vec{XY}$ ,  $\vec{YZ}$  and  $\vec{ZW}$



# Question 3

Solutions  
Student Number                     

(a)



$$\vec{OD} = \vec{OA} + \vec{AD}$$

$$= \vec{OA} + \vec{BC} \quad (\vec{AD} = \vec{BC}, \text{ opposite sides of parallelogram})$$

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} - \begin{pmatrix} -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad \text{①}$$

$$\vec{OD} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}$$

The position vector of D is  $6\mathbf{i} - 2\mathbf{j}$ . ①

(b) when Demand Price = Supply Price

$$\frac{180}{q+2} = \frac{56q}{q+1}$$

$$180q + 180 = 56q^2 + 112q$$

$$56q^2 - 68q - 180 = 0$$

$$14q^2 - 17q - 45 = 0$$

$$q = \frac{17 \pm \sqrt{17^2 - 4(14)(-45)}}{28} \quad q > 0$$

$$\therefore q = \frac{5}{2} = 2.5$$

$$\text{when } q = 2.5 \quad D = \frac{180}{4.5} = 40$$

∴ Equilibrium Price is \$40,000. ①

$$\text{Consumer surplus} = \int_0^{2.5} \left( \frac{180}{q+2} - 40 \right) dq$$

$$= \left[ 180 \ln |q+2| - 40q \right]_0^{2.5} \quad (1)$$

$$= 180 \ln 4.5 - 100 - 180 \ln 2 + 0$$

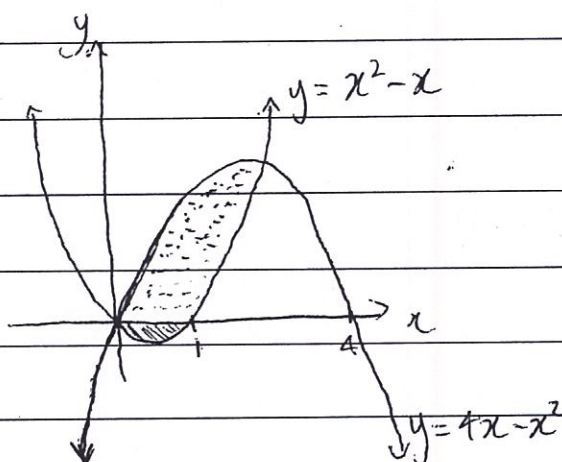
$$= 45.967$$

(1)

But  $q$  &  $D$  are in thousands so,

consumer surplus = \$4.6,000,000 to 2 significant figures. (1)

(C)



Points of intersection:

$$x^2 - x = 4x - x^2$$

$$2x^2 - 5x = 0$$

$$x(2x - 5) = 0$$

$$x = 0, 2.5$$

$$\text{Area below } x\text{-axis} = \left| \int_0^1 (x^2 - x) dx \right|$$

$$= \left| \left[ \frac{x^3}{3} - \frac{x^2}{2} \right]_0^1 \right|$$

$$= \left| \frac{1}{3} - \frac{1}{2} \right|$$

$$= \frac{1}{6} \quad (1)$$

So area above  $x$ -axis

$$= \frac{125}{24} - \frac{1}{6} = \frac{121}{24}$$

$$\therefore \text{Ratio is } \frac{1}{6} : \frac{121}{24}$$

$$= 4 : 121 \quad (1)$$

$$\text{or } 121 : 4$$

Area between curves =

$$\int_0^{2.5} (4x - x^2) - (x^2 - x) dx$$

$$= \int_0^{2.5} 5x - 2x^2 dx$$

$$= \left[ \frac{5x^2}{2} - \frac{2x^3}{3} \right]_0^{2.5}$$

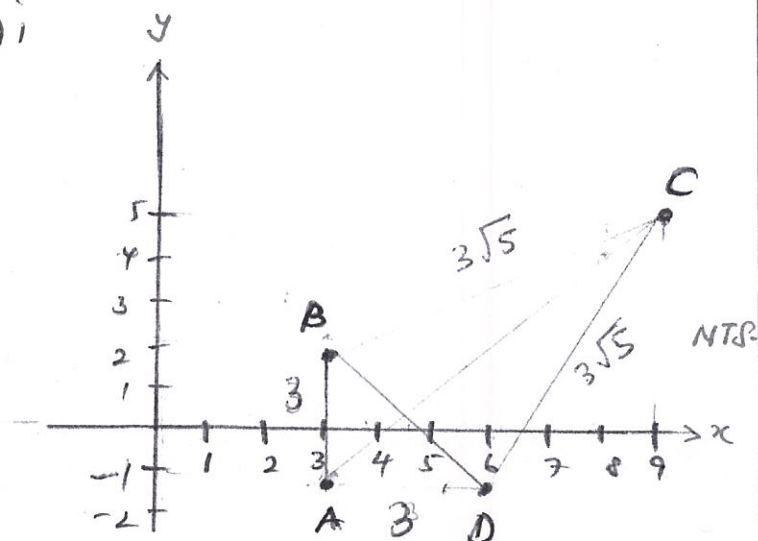
$$= \frac{125}{24} \quad (1)$$

Ext 1 4  
**MATHEMATICS: Question.....**

1/3

**Suggested Solutions****Marks Awarded****Marker's Comments**

a) i



$$\begin{aligned}\vec{AC} &= \vec{OC} - \vec{OA} \\ &= (9\hat{i} + 5\hat{j}) - (3\hat{i} - \hat{j}) \\ &= (6\hat{i} + 6\hat{j})\end{aligned}$$

$$\begin{aligned}\vec{BD} &= \vec{OD} - \vec{OB} \\ &= (6\hat{i} - \hat{j}) - (3\hat{i} + 2\hat{j}) \\ &= (3\hat{i} - 3\hat{j})\end{aligned}$$

$$\begin{aligned}\vec{AC} \cdot \vec{BD} &= 18 - 18 \\ &= 0 \quad (\text{angle is } 90^\circ)\end{aligned}$$

$$\therefore \vec{AC} \perp \vec{BD}$$

\* Always draw  
 - most students  
 were competent  
 to find  $\vec{AC}$  &  
 $\vec{BD}$  and  
 perform dot  
 product which  
 resulted in 0  
 ∴ is perpendicular

| Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks Awarded                                                                                                                                                         | Marker's Comments                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>aii) <math> \vec{AB}  =  3j </math><br/> <math>= 3</math></p> <p><math> \vec{BC}  =  6i + 3j </math><br/> <math>=  45  = 3\sqrt{5}</math></p> <p><math> \vec{CD}  =  -1 - 3i - 6j </math><br/> <math>=  45  = 3\sqrt{5}</math></p> <p><math> \vec{DA}  =  -1 - 3i </math><br/> <math>= 3</math></p> <p>adjacent sides equal and diagonals<br/>         bisect at right angle from aii)<br/>         therefore kite</p>                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>- 1 mark<br/>for each<br/>pair of<br/>correct<br/>sides</p> <p>- 1 mark<br/>for correct<br/>quadrilateral</p>                                                      | <p>if students<br/>drew the diag<br/>and labelled,<br/>would have b<br/>easy to verify</p> <p>* most students<br/>found 2 pair<br/>of equal sides<br/>but failed to<br/>determine to<br/>which family<br/>of quadrilateral<br/>ABCD belonged</p> <p>- most were<br/>able to set<br/>correctly.<br/>You cannot.</p> |
| <p>b) Prove true for <math>n=1</math></p> <p>LHS = <math>\frac{1}{1+1} = \frac{1}{2}</math>      RHS = <math>1 - \frac{1}{2} = \frac{1}{2}</math> (1)</p> <p><math>\therefore</math> LHS = RHS (statement is true for <math>n=1</math>)</p> <p>Assume true for <math>n=k</math>, where <math>k \in \mathbb{Z}^+</math>, <math>k \geq 1</math></p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <math display="block">\frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{2k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k-1} - \frac{1}{2k}</math> </div> <p>Prove true for <math>n=k+1</math></p> <p><math>\frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k-1} - \frac{1}{2k}</math></p> | <p>(2) LHS = <math>\frac{1}{2} = 1 - \frac{1}{2} = R</math></p> <p>(3) LHS = <math>\frac{1}{2}</math><br/> <math>= 1 - \frac{1}{2}</math><br/> <math>= RHS</math></p> |                                                                                                                                                                                                                                                                                                                    |

MATHEMATICS: Question.....

3/3

Suggested Solutions

Marks Awarded

Marker's Comments

LHS

$$\left( \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} \right) - \frac{1}{k+1} + \frac{1}{2k+1} + \frac{1}{2k+2}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k-1} - \frac{1}{2k} + \frac{1}{k+1} + \frac{1}{2k+1} + \frac{1}{2k+2} \quad (\text{by assumption})$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k+1} + \frac{1}{2(k+1)} - \frac{1}{k+1}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k+1} + \frac{1-2}{(k+1)}$$

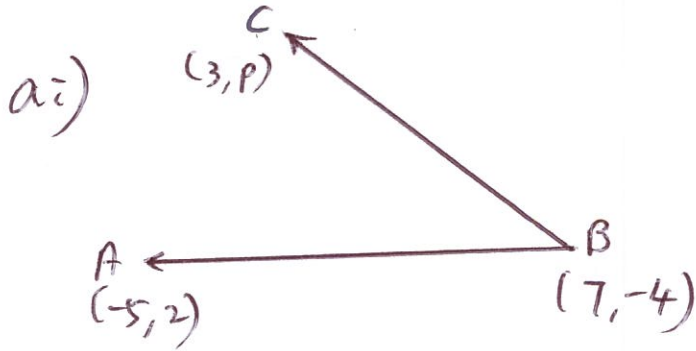
$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2k+1} - \frac{1}{2k+2} \quad *$$

= RHS

True for  $n=k$ , true for  $n=k+1$ ,  $k \geq 2 \Rightarrow k > 0$  — (induction, it  
By the process of mathematical  
is true for all positive integers

$$\begin{aligned} & -\frac{1}{k+1} + \frac{1}{2k+2} \\ & = \frac{-(2k+2) + k+1}{(2k+2)(k+1)} \\ & = \frac{-2k-2+k+1}{(2k+2)(k+1)} \\ & = \frac{-k-1}{(2k+2)(k+1)} = \frac{-1}{2(k+1)} \quad * \end{aligned}$$

many student  
couldn't simpl  
the algebraic  
part to prov  
- they knew  
what to get  
to but could  
do basic alge  
shown



FOR  $\angle ABC = 90^\circ$ ,  $\vec{BA} \cdot \vec{BC} = 0$

$$(-12\vec{i} + 6\vec{j}) \cdot (-4\vec{i} + (p+4)\vec{j}) = 0 \quad 1m$$

$$48 + 6(p+4) = 0$$

$$48 + 6p + 24 = 0 \quad 1m$$

$$6p = -72$$

$$p = \underline{-12} \quad 1m$$

i) FOR A, B, C collinear

$$\vec{BA} = k\vec{BC}$$

$$(-12\vec{i} + 6\vec{j}) = k(-4\vec{i} + (p+4)\vec{j}) \quad 1m$$

$$-12 = -4k$$

$$\therefore k = 3$$

$$6 = 3(p+4)$$

$$6 = 3p + 12$$

$$-6 = 3p$$

$$\underline{p = -2} \quad 1m$$

OR Similarly  $\vec{BA} = k\vec{AC}$  etc

well done

Wrong vectors  
max 1m

Some students  
use coordinate  
from

$$m(AC) = m(AB)$$

does not get any  
mark.

MUST show  
vectors !!

If use dot prod.  
 $\theta = 0$  or  $180^\circ$

Q5

$$b) y = 2\sqrt{x}$$

$$\frac{dy}{dx} = \frac{2}{2\sqrt{x}} = \frac{1}{\sqrt{x}}$$

1m

$$SA = \int_0^8 2\pi \cdot 2\sqrt{x} \sqrt{1 + \left(\frac{1}{\sqrt{x}}\right)^2} dx$$

$$= 4\pi \int_0^8 \sqrt{x} \sqrt{1 + \frac{1}{x}} dx$$

$$= 4\pi \int_0^8 \sqrt{x + \frac{x}{x}} dx$$

$$= 4\pi \int_0^8 \sqrt{x+1} dx$$

or

plus one  
more step

1m

$$bii) SA = 4\pi \int_0^8 \sqrt{x+1} dx$$

$$= 4\pi \frac{2}{3} \left[ (x+1)^{3/2} \right]_0^8$$

1m

$$= \frac{8\pi}{3} (27-1)$$

$$= \frac{8\pi}{3} \times 26$$

$$= \frac{208\pi}{3} \text{ units}$$

1m

P 3/2

without clearly  
showing  $\frac{dy}{dx} = \frac{1}{\sqrt{x}}$

$$2\pi \cdot 2\sqrt{x} \sqrt{1 + \left(\frac{1}{\sqrt{x}}\right)^2}$$

max only 1m

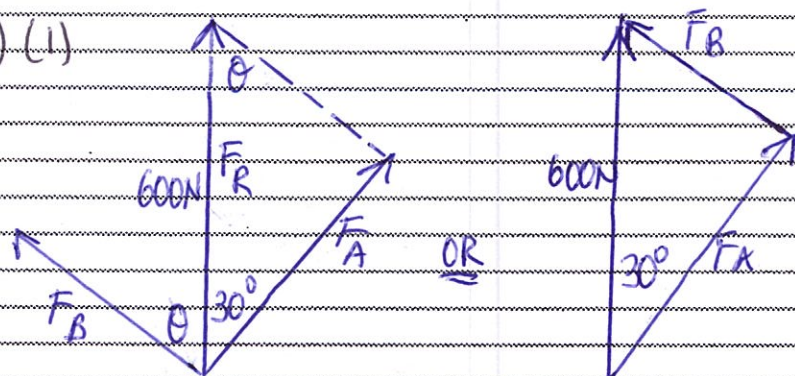
MATHEMATICS Extension 1 : Question...7

Suggested Solutions

Marks

Marker's Comments

a) (i)



$F_B$  is minimum when the angle between  $F_A$  and  $F_B$  is  $90^\circ$  as the shortest distance is the perpendicular distance.  $\therefore \theta = 60^\circ$

$$(ii) F_A = 600 \cos 30^\circ$$

$$= 600 \times \sqrt{3}/2$$

$$= 300\sqrt{3} \text{ N}$$

$$F_B = 600 \sin 30^\circ$$

$$= 300 \text{ N}$$

$$b)(i) \text{ Volume} = \pi \int_1^2 y^2 dx$$

$$= \pi \int_1^2 x^4 + 2x^2 + 1 dx$$

$$= \pi \left[ \frac{x^5}{5} + \frac{2x^3}{3} + x \right]_1^2$$

$$= \pi \left[ \frac{32}{5} + \frac{16}{3} + 2 - \frac{1}{5} - \frac{2}{3} - 1 \right]$$

$$= \frac{178\pi}{15} \text{ cubic units}$$

Diagrams were poorly drawn, most students did not show 600N along y-axis. Students need to understand that the resultant force is the sum of  $F_A$  and  $F_B$ . Students did not explain that the shortest distance is perpendicular distance.

Most errors were made in the substitution. Students who showed the substitution were most successful.

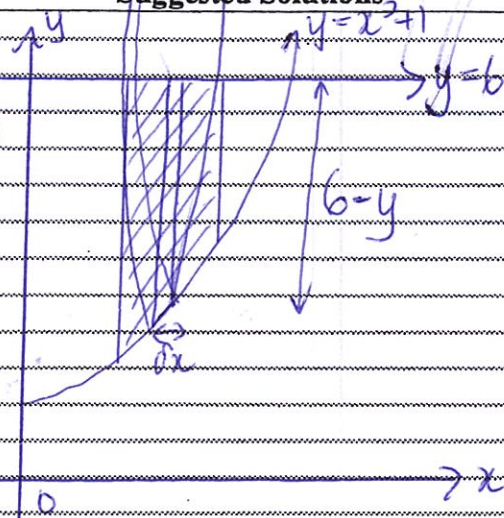
Suggested Solutions

Marks

Marker's Comments

b) ii)

Method 1:



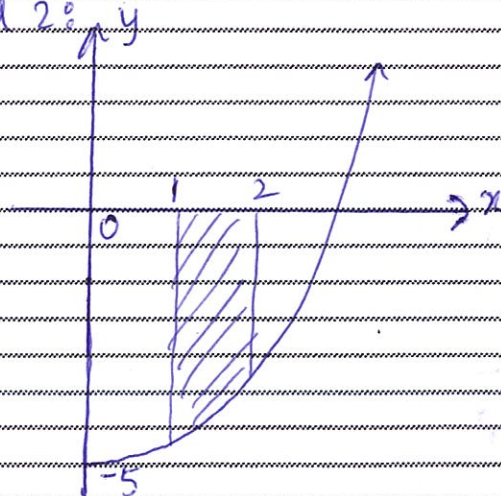
Rather than rotate the curve about  $y=0$  with radius of the disc being  $y$ , the radius of the disc is  $6-y$ .

We then sum up the disc volumes to give

$$V = \pi \int_1^5 (6-y)^2 dx = \pi \int_1^5 (5-x^2)^2 dx$$

$$= \pi \int_1^5 x^4 - 10x^2 + 25 dx$$

Method 2:



Volume is equivalent to the volume of  $y = x^2 + 1$  shifted down 5 units and rotated about  $y=0$ .  
i.e.  $y = (x^2 + 1) - 6 = x^2 - 5$

$$V = \pi \int_1^2 (x^2 - 5)^2 dx = \pi \int_1^2 x^4 - 10x^2 + 25 dx$$

1 → for diagram showing radius

1 → for explanation

1 → Volume expression which follows from diagram and explanation

1 → diagram showing shift

1 → explanation of shift,  $y = x^2 - 5$  and  $y=0$  as axis of rotation

1 → expression which follows from

explanation